

Exploring the Usability of African-Inspired Wearable Cameras Co-designed Through a Conflict Sensitive Design Approach

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Abstract

Body-worn cameras (BWCs) are widely promoted to improve accountability, trust, and transparency in community policing. However, most existing BWC designs and deployment models largely reflect assumptions from the Global North and often overlook the cultural, infrastructural, and socio-political realities of African contexts, where relationships between police and citizens are frequently adversarial. This has affected the use of BWCs in community policing engagements in Nigeria, as existing designs do not account for the power imbalance among stakeholders. In a preliminary study, a conflict-sensitive design approach was adopted to co-design BWCs with Nigerian police officers and citizens, resulting in African-inspired BWC prototypes, that reflected their locally grounded design choices. A one-week diary study with 12 participants wearing the prototypes, was conducted across five scenarios in this study, along with focus group interviews. Findings reveal diverse device preferences, usage practices, and behavioral responses, highlighting locally aligned design opportunities for community policing technology designers.

CCS Concepts

• Human-centered computing → Usability testing.

Keywords

Community policing, Technology, Usability Testing, Citizens, Police

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1 Introduction

Body-worn cameras (BWCs) continue to gain global attention as tools for enhancing accountability, trust, and transparency in policing[29].

Yet, much of their design and deployment has been shaped by sociotechnical assumptions rooted in Global North contexts, with limited consideration for cultural norms, infrastructural constraints, and community dynamics in African context.

Community policing (CP) is a policing strategy where the citizens work with the police to address security concerns[26]. Hence, the police and citizens are important stakeholders of CP. In Nigeria, where insecurity and strained police–citizen relations are longstanding concerns[9], the introduction of wearable technologies for community policing demands contextually grounded and conflict-sensitive design approaches [25, 28]. Building on insights generated through preliminary research [27] with Nigerian community policing stakeholders, we developed a set of low-fidelity, African-inspired body-worn camera prototypes; integrated into hats, wristbands, and cloth buttons, to probe wearable camera concepts. Because of the discrepant preferences identified in a previous study [28], we were particularly interested in the usage modalities and visibility options of the BWCs in comparison with practices in the Global North. Hence, the prototypes incorporated voice, gesture, and touch activation modalities and were developed in both hidden and visible forms to reflect divergent preferences regarding discretion, security, and social signaling. To investigate stakeholders’ experiential interactions with these prototypes, evaluate their potential adoption, and explore further design opportunities for community policing, a usability study was conducted, using a diary-based method, complemented by focus group interviews. This approach enabled the capturing of nuanced ways in which context, social dynamics, and cultural expectations shape the use and perception of wearable cameras.

1.1 Conflict Sensitive design

Conflict Sensitive Design (CSD) is a six-phase framework that enables adversarial stakeholders to participate in design activities together, but from different locations and at different times [27]. The six phases include: Identify stakeholders and form teams of participants, requirements gathering, design with marginalized users, design with Up users (authority group), iterate until design agreement is reached, and evaluate the designs and prototype. Amid these phases are five mediation techniques of separated meetings, tension reduction, leveling, common ground reminder and formalizing agreement. An initial study [28] found that Global North-inspired (unhidden) BWCs were unpreferred by Nigerian citizens and identified adversarial stakeholders (Nigerian citizens and police) who



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were unwilling to participate in design activities together due to power asymmetries. In response, the CSD framework was developed [27] to address the challenges of co-designing with these groups. The present study represents the sixth and final phase of the CSD framework, evaluating the designs and prototype. This work explores stakeholders' experiences and derive design implications for the adoption of community policing technologies in Nigeria.

The contributions of this study include:

- (1) Providing empirical evidence that shows how cultural, security, and socio-political factors in Nigeria shape preferences and use of BWCs, challenging Global Northern assumptions and expanding HCI and policing literature with insights grounded in an African context.
- (2) Evaluating novel BWC form factors (wrist-worn, cloth-button, and head-worn), and offering concrete design opportunities for community policing technologies tailored to African settings.

2 Related works

2.1 Body-Worn Cameras in diverse contexts

Body-worn cameras (BWCs) have been widely adopted across policing agencies, particularly in the Global North countries as tools for enhancing accountability, transparency, and evidentiary reliability [2, 16]; and have been associated with reductions in use-of-force incidents [10, 17]. Furthermore, studies assume strong acceptance of use of BWCs by relevant stakeholders in China, Germany, USA and UK [3, 18]; conditions that differ considerably from Nigerian contexts [28]. Consequently, the applicability of these findings to Nigeria, where police–citizen relations are historically strained [13] and public safety infrastructures are fragmented [9], requires careful interrogation.

2.2 Cultural Contexts in Wearable Camera Adoption and Use

Studies on wearable cameras have explored privacy perceptions and usage contexts, yet cultural considerations remain underexamined. Singhal et al. [31] studied bystanders' perceptions of privacy infringements in public spaces, offering insights into general privacy expectations but overlooking how cultural norms and community values influence the acceptance of body-worn cameras. Similarly, Brush et al. [7] examined a Digital Neighborhood Watch system and found willingness to share camera footage for evidentiary purposes, without addressing how community sharing norms may vary across cultural settings. Yang and Shihada [35] argue that cultural acceptability plays a critical role in wearable design, underscoring the need for context-sensitive evaluations. These inform the present study, which explores wearable camera usability through scenario-based methods in the Nigerian context.

3 Methods

Building on the design outcome of a previous CSD co-design workshops [27], the author developed body-worn camera prototypes by repurposing Amazon Insta360 GO 3S cameras with voice, gesture and touch activation functionalities, and customizing them into African-inspired

wearable form; hats, wrist bangles, and cloth buttons. For each prototype with a hidden camera concealed in a sticky closure, there was a corresponding prototype with an unhidden camera, and both were provided to the participants. We conducted a usability study using a diary study approach [20], employing the experience prototypes.

4 Recruitment and Participants

A total of 12 stakeholders (6 citizens: 5M, 1F, ages 18–60 and 6 police officers: 6M, ages 39–59) participated in the diary study. These participants were recruited from an earlier co-design workshop. We subscribed 10GB worth of data bundle (\$5) for each of the participants and gave them N30,000 which is the equivalent of \$20. This study received ethical approval from the relevant ethics committee of my school.

5 Data Collection

5.1 Diary Study

One week diary study was conducted by the author, employing the elicitation study approach [22, 34]. During the study briefing, both citizens and police participants were instructed on how to activate or deactivate the prototypes using three interaction modalities: voice (by saying “start recording” or “stop recording”), gesture (by waving over the device), and touch (by pressing the button), depending on preferences at shopping malls, offices, churches, crowded areas, and own scenarios. Consistent with the approach used in [11], the devices' recording functions were disabled without the participants' knowledge. At the debriefing stage, participants were told that the devices, despite showing activation and deactivation indicators when operated, did not perform any real recording. After the diary study, focus group interviews [15, 24] were conducted to elicit detailed insights from participants regarding their experiences, perceptions, and challenges with the use of the prototypes.

Seventy seven diary entries were submitted by the diarists via WhatsApp as suggested in the study by Li et al. [22]. The diary entries which contained their daily experiences with the devices and modality of use acted as prompts which enabled the researcher and participants to revisit and reflect on their recorded experiences, thereby guiding the focus group interviews specifically.

5.2 Focus Group Interview

To enable open discussions and capture a wider range of participant perspectives regarding their experiences with the prototypes, as recommended by Lazar et al. [21], two separate focus groups were conducted by the author and two other researchers, following conflict sensitive approach [27]; one with citizens and one with police officers. Each focus group session lasted a minimum of 120 minutes. A total of 14 questions were posed in each focus group with prompts on feelings of comfort and ease of use, usefulness, change in behavior, cultural perception, hidden or unhidden device preferences, modality of use, privacy and confidentiality.

6 Data Analysis

Data analysis was conducted as a collaborative group activity involving the author and two other researchers. Following Braun

and Clarke's six-phase reflexive thematic analysis [6], the author and one researcher independently familiarized themselves with the transcripts and generated initial codes. Theme development was then carried out collaboratively with the second researcher, using an established HCI approach [19]. During this phase, the author led the writing of integrative memos to clarify categories, explore relationships, and identify subthemes. The first and second researchers iteratively reviewed and refined the themes to ensure quality, coherence, and clear boundaries, consistent with prior HCI studies [12, 14]. A third researcher with expertise in qualitative analysis validated the final themes. Participant validation was subsequently conducted via WhatsApp following Birt et al. [5].

7 Findings

The study uncovered key insights into device preferences, usage patterns, and behaviors related to wearable cameras among Nigerian community policing stakeholders. The themes highlighted the complexities of technology adoption and revealed new design opportunities for community policing technologies. We present these themes using identifiers P1 to P6 for citizen participants and PP1 to PP6 for police participants.

7.1 User comfort with concealable devices

Citizens and police officers expressed a clear preference for hidden body-worn cameras (BWCs) over visible ones, as they allowed participants to record their surroundings without provoking potential aggression. One citizen participant (P1) stated, "Hidden one was always better," while a police officer (PP1) noted, "I felt comfortable, most especially while using the hidden devices." This comfort is particularly relevant in the context of Nigeria, where insecurity heightens the perceived need for protective technologies. Ergonomics and simplicity were also stressed to accommodate older adults and women, with P2 highlighting, *"It should be lightweight and small. Older adults or women might struggle with bulky designs."*

7.2 Preferred interaction modalities

Feedback indicated that devices with more intuitive interaction modalities, particularly those utilizing voice commands and simple gestures enhanced the overall user experience and engagement. Participants easily activated the cameras with voice commands and hand gestures in all scenarios with only one (PP6) touching the screen in own scenario. Participants found it difficult to locate the touch button (lens), especially for the hat camera. Also, they felt it was hard to bend their necks to trace the buttons of the cloth-button and wrist-worn cameras before touching them. According to P1 *"finding the activation button was hard, so I just wave to it to start and wave back to stop"*. Different contexts, such as crowded places, and personal encounters, presented varying challenges and opportunities for the use of wearable devices. For instance, a citizen participant (P3) was distracted in religious spaces and *"wasn't comfortable activating this in the church"*.

8 Discussion

The findings yield significant insights into implications for technology adoption in local contexts.

Firstly, participants generally perceived the BWCs, particularly the concealed recording devices, as comfortable, acceptable, and usable. This contrasts with Global North studies by Singhal et al. [32], Brush et al. [8], Bhardwaj et al. [4], and Alharbi et al. [1], which report discomfort and ethical challenges associated with the use of hidden body-worn cameras (BWCs). Participants' comfort in our study appears to reflect the high levels of insecurity in Nigeria, which may have led them to prioritize personal safety over ethical considerations. Additionally, because negative police behavior was not the primary focus of the study, participants may have experienced less discomfort. Notably, police officers felt empowered using these devices, which they associated with a means of ensuring their safety while on duty. This indicates that technology design and deployment must be sensitive to local perceptions of safety and risk rather than relying on assumptions derived from foreign contexts.

Secondly, the appearance and modalities of interactions of the body-worn camera prototypes play an important role in the experience of users and their willingness to use the devices for community policing. Devices with easy interaction modalities of voice or gesture enhanced usability and acceptance to engage in community policing. This is consistent with the findings of Burst et al. [23]. However, participants' preference for voice and gesture-based activation contrasts with Global North BWC form factors, where police officers typically conspicuously activate BWCs by manually pressing buttons [30, 33], often without concern for risks associated with visibility as it is in Nigerian context.

9 Design Implications

The findings of this study suggest several important design implications:

1) BWC designs should explicitly account for high-insecurity environments where visibility can expose users to harm. Concealed or discreet form factors are critical in such contexts, as they align with users' prioritization of personal safety over deterrence and visibility-driven transparency norms common in Global North deployments.

2) Interaction techniques that minimize overt physical actions, such as voice and gesture-based controls, should be favored in contexts where conspicuous device manipulation may escalate tension or draw unwanted attention to armed criminals.

3) In high-risk settings, ethical design should accommodate contextual trade-offs where concealment may be necessary for safety, while still ensuring appropriate governance, oversight, and data protection mechanisms. In specific sociopolitical contexts, covert recording might be ethically justified as a protective measure. Embedding cameras into everyday accessories or garments can reduce stigma, increase usability, and facilitate smoother community policing interactions.

10 Conclusion

Where Western studies primarily frame body-worn cameras through lenses of transparency and accountability, this study reframes them as tools for safety and survival in contexts shaped by distinct insecurity challenges and cultural realities. Participants' narratives reveal that technological adoption is relational and deeply embedded in local conditions. At the same time, the introduction of discreet or

hidden BWCs carries risks of misuse, underscoring the need for context-sensitive ethical frameworks and transparent policies that address power disparities and prevent surveillance from becoming a tool of oppression rather than protection. While deeper analysis is reserved for future work, this study addresses a significant gap in literature dominated by Global North assumptions, demonstrating that discreet and intuitive BWCs can support community policing engagement in local contexts.

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